Cardiac Companion

An offline medicine record for young people with rheumatic heart disease in Ethiopia

Team: TODO (name, school). Repo: https://github.com/aminmohammed2k7-lgtm/cardiac-companion Demo: https://cardiac-companion.vercel.app (tap "Try the sample record")

The problem

Young people with rheumatic heart disease often take warfarin, which needs regular INR blood tests, and a penicillin injection every few weeks. Missing either can be serious. Most track this on a paper card.

What it does

- Shows today's warfarin dose as the doctor wrote it. It never changes a dose.
- Keeps INR results against the doctor's target range.
- Counts down to the next penicillin injection.
- Lists medicines by time and warns before tablets run out.
- Lists warning signs with one-tap calls to the clinic and family.
- Makes a visit summary PDF for the doctor.
- Estimates pulse with the phone camera (an estimate, not a medical device).

How people use it

Open the app, pick a language (English, Amharic or Afaan Oromo), enter the doctor's instructions, then check Today each day. A sample record and a short guided tour show everything without typing.

How it is built

Android app with Capacitor, plain HTML and JavaScript. Works offline. No servers or accounts: data stays on the phone, with encrypted backup and an optional PIN. 75 automated test runs check the medical rules in three time zones, all three languages and the design rules.

Limits and next steps

Camera pulse accuracy is not yet tested against a pulse oximeter. Local language text needs a native speaker check. Next: phone testing, the accuracy check, and a clinician review of the wording.